

MERIDIAN VALLEY DIAGNOSTICS — SYNTHETIC TEST DOCUMENT

Patient: TESTCASE, Alex (fictional) · DOB 1980-03-11 · Sex M · MRN SYN-000-114 · Collected 2026-02-14 · Reported 2026-02-16

Comprehensive metabolic and lipid panel

Analyte	Result	Unit	Reference interval
Total protein	68	g/L	64–83
Albumin	41	g/L	35–50
Glucose, fasting	6.4	mmol/L	3.9–5.6
Haemoglobin A1c	42	mmol/mol	20–41
Ferritin	312	µg/L	30–400
Creatinine	94	µmol/L	62–106
eGFR (CKD-EPI)	88	mL/min/1.73m ²	> 90
Total cholesterol	5.8	mmol/L	< 5.2
LDL cholesterol	3.9	mmol/L	< 3.0
HDL cholesterol	1.1	mmol/L	> 1.0
Triglycerides	2.1	mmol/L	< 1.7
25-OH vitamin D	48	nmol/L	50–125
TSH	2.4	mIU/L	0.4–4.0
ALT	46	U/L	10–40
C-reactive protein	4.8	mg/L	< 5.0

Imaging — coronary CT angiography, 2026-02-09

Coronary artery calcium score 212 (Agatston), 78th percentile for age and sex. Mixed plaque in the proximal LAD with an estimated 30–40% diameter stenosis. Right coronary artery and left circumflex without haemodynamically significant disease. No anomalous origin.

Comparison: prior calcium score 168 on 2024-01-22 — interval progression.

Impression

- Impaired fasting glucose with HbA1c in the prediabetic range.
- Mixed dyslipidaemia, LDL above target for calculated risk.
- Subclinical coronary atherosclerosis, progressive by calcium score.
- Vitamin D insufficiency. Mildly elevated ALT, unexplained.

Synthetic document generated for benchmarking. Not a real patient, not a real laboratory, not for clinical use.